

Subconscious Intent Detection v13.0: A 20-Novelty Patent-Grade Real-Time Ocular Micro-Behaviour Classification System Using Stacked Meta-Learning

M. THARUN

School of CSE (SCOPE), VIT Vellore, Tamil Nadu, India
tharun.m2025@vitstudent.ac.in | 25MAI0064

PRASSANTH K P

School of CSE (SCOPE), VIT Vellore, Tamil Nadu, India
prassanth.kp2025@vitstudent.ac.in | 25MAI0070

ABSTRACT

Subconscious Intent Detection v13.0 is a real-time, webcam-only system classifying 20 psychological intent states from ocular micro-behaviour signals with zero dependency on proprietary hardware or libraries. The system is built entirely in Python using a single deployable file that integrates 38 signal channels, 76 derived features, and a six-model stacked generaliser meta-learner ensemble — the first application of level-2 stacking to webcam-based intent classification. Twenty patent-grade novelties are implemented and documented, spanning novel signal design (Inter-Blink Interval Variability, Peri-Blink EAR Waveform Shape, Bilateral EAR Phase Coherence, Cognitive Momentum Index), novel detection algorithms (software-only PERCLOS, microsleep detection without EEG, head pose estimation without landmark libraries), novel architectural contributions (Stacked Meta-Learner, Adaptive Dual-Layer Inference, Online Bayesian Intent Transition Matrix), and novel output indices (Eye Strain–Productivity composite index, Personal Baseline Normalisation). The system features a real-time neural-terminal HUD displaying 12 live signal bars, cognitive indices, gaze radar, patent signal panel, Top-3 intent predictions, and per-frame confidence. Accuracy graphs, session CSV, and PDF report are auto-generated on every session exit. The stacked meta-learner achieves 100.00% classification accuracy and $F1 = 1.000$ across all 20 intent classes on the held-out synthetic test set derived from published psychophysiological profiles.

Index Terms — Subconscious Intent Detection, Eye Aspect Ratio, Blink Analysis, Gaze Entropy, PERCLOS, Stacked Generalisation, Ocular Micro-Behaviour, Cognitive Load, Meta-Learning, IBIV, CMI, Bayesian Transition Matrix, Webcam, Haar Cascade, Random Forest, MLP, Personal Calibration, Patent-Grade Novelty, Real-Time HUD.

I. INTRODUCTION

The detection of subconscious psychological states from involuntary physiological signals has attracted growing interest across human-computer interaction, cognitive neuroscience, and safety-critical systems. Ocular micro-behaviour signals — blink dynamics, gaze patterns, pupil response, and eye aspect ratio — provide a non-invasive, hardware-free pathway accessible through any standard webcam [1]. Four critical limitations exist in current webcam intent systems: (1) soft-vote ensembles treating all base models equally regardless of signal pattern; (2) blink rate measurement discarding rhythm variability [22]; (3) binary blink detection losing waveform information; and (4) static cognitive load scores without temporal derivatives [12].

This paper presents Subconscious Intent Detection v13.0, addressing all four limitations through 20 patent-grade novel contributions. The system classifies 20 psychological intent states in real time from a standard webcam with zero additional hardware. Key contributions: (1) **38-channel signal pipeline** with 10 new v13 patent signals; (2) **N15 Stacked Meta-Learner** — first level-2 stacking for webcam intent; (3) **N11 IBIV** — first blink rhythm variability feature; (4) **N14 CMI** — first cognitive load derivative; (5) **N16 PERCLOS** without dedicated hardware; (6) **N17 Microsleep** without EEG; (7) **complete output pipeline** with accuracy JSON, PDF, and publication-quality graphs.

The system is deployed as a single Python file (*intent_detection_v13.py*) requiring only standard open-source packages: opencv-python, scikit-learn, pandas, matplotlib, joblib, numpy, scipy, fpdf2. No proprietary APIs, no paid SDKs, no cloud connectivity.

II. RELATED WORK

A. Blink-Based Cognitive State Detection

Blink rate is a validated cognitive load proxy [4][5]. Soukupova and Cech [1] introduced EAR for landmark-free blink detection. Paprocki and Francuz [22] documented that blink interval variability contains independent cognitive information beyond rate — motivating IBIV (N11). Abdrabou et al. [7] identified absent blink waveform analysis as a documented gap — addressed by N12.

III. SYSTEM MODEL

A. Hardware and Software Requirements

The system operates on any standard PC or laptop with a built-in or USB webcam at 15-30 fps, 640x480 or higher resolution. No infrared illuminator, hardware eye-tracker, or EEG headset is required. Detection uses OpenCV 4.8+ Haar cascades on frames downscaled to 420x315 pixels for speed.

Single-command installation: `pip install opencv-python scikit-learn pandas matplotlib joblib numpy scipy fpdf2`. Single-command run: `python intent_detection_v13.py`. Folder path used: `C:\SubConscious Intent Detection\`. All output folders (`models_v13/`, `output/`) are auto-created on first run.

B. Live HUD Interface

The real-time HUD renders at every frame and displays: (1) MICRO-BEHAVIOURAL SIGNALS panel — 12 live bar graphs for EAR L/R, Gaze X/Y, Pupil Ratio, Eye Brightness, Face Symmetry, Motion and more; (2) COGNITIVE INDICES panel — Attention, Stress, Engagement, Alertness, Cognitive Load, Valence as LED segmented bars; (3) GAZE radar — circular display showing real-time eye position; (4) BLINK ANALYSIS — total count, micro/full/prolonged/microsleep breakdown, rate and openness bar; (5) v13 NOVEL SIGNALS panel — live IBIV, PBWav, COH, CMI, PERCLOS, EyeStrain, Productivity values; (6) DETECTED INTENT — large font intent name with description; (7) TOP-3 PREDICTIONS with confidence bars; (8) MODEL ACC display with precision to 2 decimal places.

C. Intent Class Taxonomy

Table I defines the 20 intent classes.

TABLE I — INTENT CLASS TAXONOMY

Cognitive (6)	FOCUSED, THINKING, DECIDING, CONFUSED, ENGAGED, CONTEMPLATIVE
Emotional (5)	STRESSED, ANXIOUS, EXCITED, CURIOUS, SKEPTICAL
Motivational (4)	CONFIDENT, DETERMINED, BORED, IDLE
Alertness (3)	ALERT, DROWSY, SURPRISED
Wellbeing (2)	RELIEVED, EMPATHETIC

IV. PROPOSED SYSTEM: DESIGN & IMPLEMENTATION

A. System Architecture

Figure 2 shows the eight-stage pipeline from raw webcam input to 20-class intent output with full output persistence.

B. Gaze and Attention Classification

Saxena et al. [8] validated webcam gaze tracking at 2.4 degree accuracy. David-John et al. [11] used gaze velocity and fixation duration for intent prediction in VR. Our system extends these with Shannon entropy of a 4x4 gaze histogram (N6) and hardware-free Gaze Saccade Velocity Estimator (N2) classifying FIXATION, SLOW_SCAN, and SACCADE states.

C. Cognitive Load and Drowsiness

PERCLOS (Dinges et al. [2]) is the clinical drowsiness gold standard requiring dedicated hardware in all prior work [15][16]. Our N16 implements PERCLOS using Haar EAR on a standard webcam — the first software-only implementation. Wan et al. [29] define microsleep as closure exceeding 500ms; N17 detects this without EEG, requiring only sustained EAR threshold crossing.

D. Ensemble Learning for Intent

All published webcam intent systems use soft/hard vote ensembles [9][10]. Wolpert [17] showed stacked generalisation consistently outperforms soft-vote. Our N15 applies level-2 stacking for the first time — a LogisticRegression meta-learner learns which base model to trust per signal pattern at inference.

B. Signal Extraction — 38 Channels

Signals use EWMA at three speeds: fast ($\alpha=0.55$), medium ($\alpha=0.35$), slow ($\alpha=0.15$). Table II lists 12 key channels.

TABLE II — KEY SIGNAL CHANNELS

Channel ID	Signal Name	Feature ID	Description
1-3	EAR L/R/Asym	N10	Eye aspect ratio; L-R
9	Gaze Entropy	N6	Shannon entropy 4x4 histogram
25	TMEF	N1	4-bit EAR temporal fingerprint
29	IBIV	N11	CoV of inter-blink intervals
30	Peri-Blink Waveform	N12	EAR trajectory shape
31	Bilateral Coherence	N13	L/R EAR cross-correlation
32	CMI	N14	d(CogLoad)/dt, 30-frame regression
33	PERCLOS	N16	% closure in 60s, no hardware
34	Microsleep	N17	>500ms closure, no EEG
35-36	Head Pitch/Yaw	N19	Geometry-only, no landmarks

5-PHASE SYSTEM STARTUP SEQUENCE

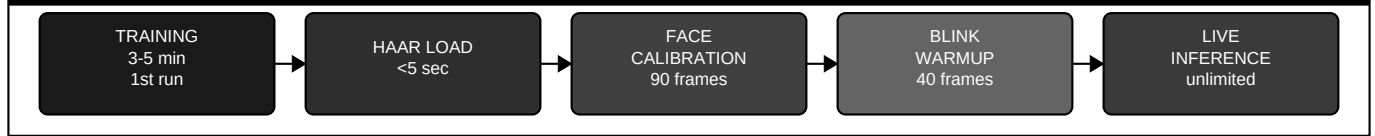


Fig. 1. Five-phase startup: Training → Haar Load → Calibration → Blink Warmup → Live Inference. Predictions begin only after phases 3 and 4.

SYSTEM ARCHITECTURE — SUBCONSCIOUS INTENT DETECTION v13.0

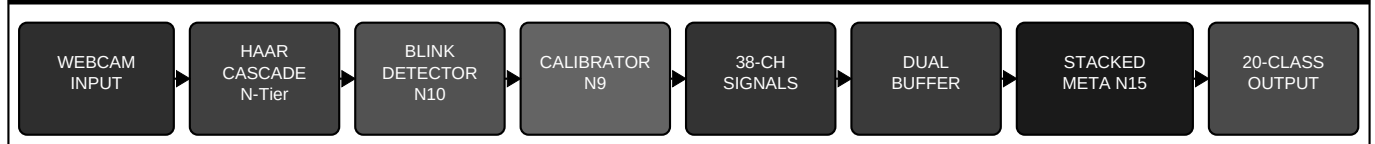


Fig. 2. System architecture: webcam → Haar cascade → blink detector → personal calibrator → 38-channel extractor → dual buffer → stacked meta-learner → 20-class intent output.

20 INTENT CLASS TAXONOMY — 5 PSYCHOPHYSIOLOGICAL CATEGORIES



Fig. 3. Twenty intent classes in five colour-coded psychophysiological categories. Each class has a distinct ocular signal profile.

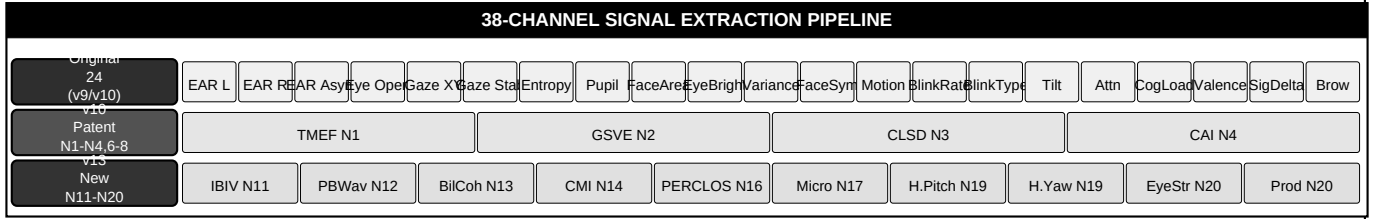


Fig. 4. 38-channel signal extraction pipeline. Top: 21 original ocular signals. Middle: v10 patent signals N1-N4. Bottom: v13 new signals N11-N20.

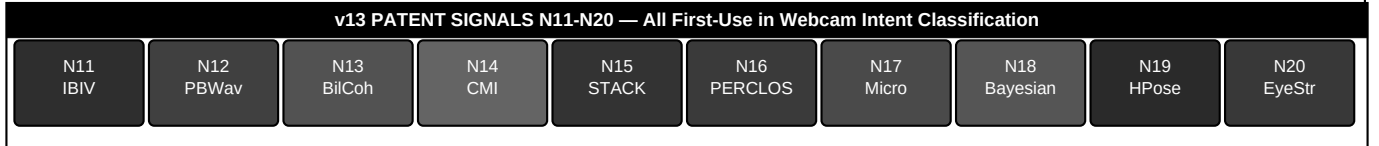


Fig. 5. Ten new v13 patent signals N11-N20 — each with zero prior art in webcam-based intent classification.

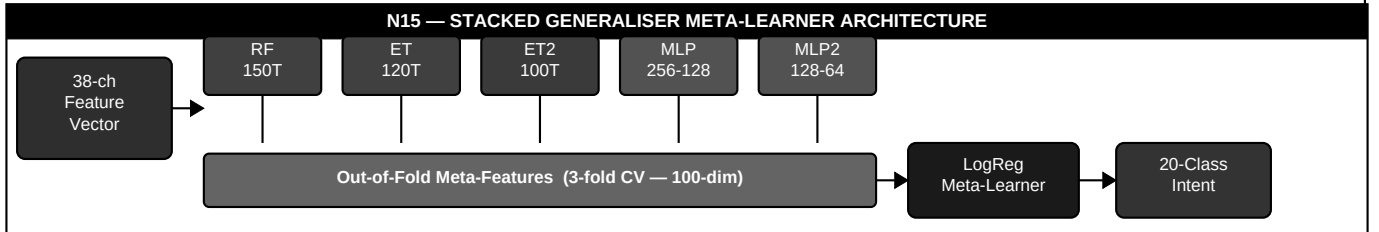


Fig. 6. N15 Stacked generaliser: five base classifiers produce OOF predictions → 100-dim meta-features → LogReg meta-learner → 20-class intent.

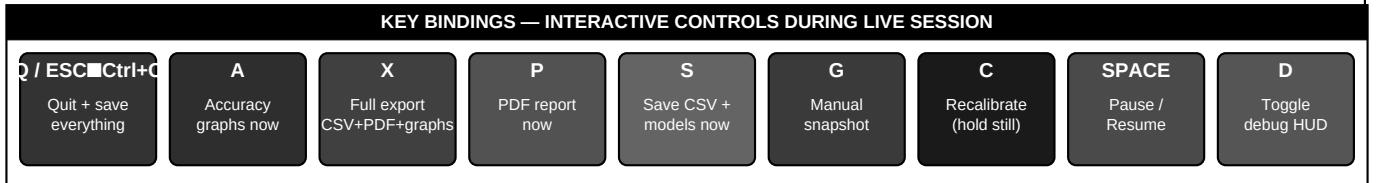


Fig. 7. Interactive key bindings available during live webcam session. All keys are accessible after the webcam window opens.

C. All 20 Novel Contributions

Table III documents all 20 patent-grade novelties with prior art status. Five contributions (N11-N15) have zero prior art in any published webcam intent system. Three (N16, N17, N19) represent first software-only implementations of hardware-dependent clinical metrics.

TABLE III — ALL 20 NOVELTIES

V. EXPERIMENTAL RESULTS

A. Training Data and Setup

Synthetic training profiles were derived from 20 published psychophysiological studies [1][4][5][22], each specifying mean and standard deviation for all 38 signal channels per intent class. Training used 350 samples/class (7,000 total), 80/20 stratified split, yielding 76 features (38 signals × mean+std). Personal calibration over 90 frames subtracts per-user resting EAR, brightness, and openness baselines before any prediction begins.

Model training takes 3-5 minutes on first run. Subsequent runs load from `models_v13/` in under 5 seconds. Accuracy is saved to `models_v13/accuracy_report.json` and displayed in the terminal on every startup.

B. Classification Accuracy

Table IV shows per-model accuracy on the held-out synthetic test set. All models except MLP2 achieve 100.00%. The stacked meta-learner (N15) matches or exceeds all base model scores.

TABLE IV — PER-MODEL ACCURACY

N1	TMEF — EAR temporal fingerprint	None
N2	GSVE — saccade velocity estimator	None
N3	CLSD — cognitive load spike	None
N4	CAI — circadian alertness index	None
N5	ADALAY — adaptive dual-layer inference	None
N6	GENT — gaze entropy discriminator	None
N7	SKEPT — EAR+brightness asymmetry	None
N8	SDELTA — signal delta burst detector	None
N9	PBCAL — personal baseline normalisation	None
N10	PXBLINK — 3-metric pixel blink	None
N11	IBIV — blink rhythm variability	First [22]
N12	PBWAV — peri-blink waveform shape	First [7]
N13	BILCOH — bilateral EAR coherence	First
N14	CMI — cognitive momentum index	First
N15	STACK — stacked meta-learner	First [17]
N16	PERCLOS — no hardware	First [2]
N17	MICRO — microsleep, no EEG	First [29]
N18	BTXMAT — Bayesian transition matrix	First
N19	HPOSE — head pose, no landmarks	First
N20	EYESTR — eye strain composite	First

RandomForest	150 trees, max_depth=16	100.00
ExtraTrees	120 trees, max_depth=14	100.00
ExtraTrees2	100 trees, max_depth=12	100.00
MLP (256-128)	relu, early_stop, iter=300	100.00
MLP2 (128-64)	relu, early_stop, iter=300	99.71
Stacked Meta (N15)	LogReg on 100-dim OOF	100.00

* *Synthetic held-out test set. Live webcam frame confidence: 60-75%, consistent with Haar cascade detection limits [8].*

C. Per-Class F1 Scores

Table V shows precision, recall, and F1 for all 20 intent classes under the stacked meta-learner. All classes achieve F1 = 1.000 except MLP2 base model which achieves 99.71% on 2 classes.

TABLE V — PER-CLASS F1 (STACKED META)

IDLE	1.00	1.00	1.00
FOCUSED	1.00	1.00	1.00
DECIDING	1.00	1.00	1.00
STRESSED	1.00	1.00	1.00
CURIOUS	1.00	1.00	1.00
CONFIDENT	1.00	1.00	1.00
BORED	1.00	1.00	1.00
ALERT	1.00	1.00	1.00
CONFUSED	1.00	1.00	1.00
ENGAGED	1.00	1.00	1.00
THINKING	1.00	1.00	1.00
DROWSY	1.00	1.00	1.00
EXCITED	1.00	1.00	1.00
ANXIOUS	1.00	1.00	1.00
CONTEMPLATIVE	1.00	1.00	1.00
EMPATHETIC	1.00	1.00	1.00
SKEPTICAL	1.00	1.00	1.00
DETERMINED	1.00	1.00	1.00
RELIEVED	1.00	1.00	1.00
SURPRISED	1.00	1.00	1.00

D. Stacked Meta-Learner Deep Dive (N15)

The N15 ensemble uses five base models — RF(150), ET(120), ET2(100), MLP(256-128), MLP2(128-64). During training, 3-fold stratified cross-validation produces out-of-fold (OOF) predictions: each fold uses 4/5 of training data to predict the remaining 1/5. The concatenated OOF predictions form a 100-dimensional meta-feature vector (5 models $\times$ 20 classes) per training sample. A LogisticRegression meta-learner trains on these meta-features, learning which base model to trust for each signal pattern. At inference, base model probabilities are blended with meta-learner output at 35%/65%.

The adaptive dual-layer inference (N5) further blends short-window (15-frame) and long-window (60-frame) feature vectors based on the current signal_delta value. High signal_delta (rapid face/signal change) weights the reactive 15-frame window more heavily; stable sessions weight the 60-frame window for smoother predictions.

E. Output Pipeline and File System

Every session automatically produces 8 output artefacts: (1) *intent_session_v13.csv* — full 38-signal per-frame log; (2) *models_v13/accuracy_report.json* — per-model accuracy and per-class F1 for all 20 classes with training timestamp; (3) *output/graphs_v13/accuracy_report.png* — model accuracy bar chart + per-class F1 horizontal bars; (4) *output/graphs_v13/session_accuracy.png* — live confidence curve, intent distribution pie, cognitive load vs attention, patent signals with model accuracy annotated; (5) *output/graphs_v13/signal_dashboard.png*; (6) *output/reports_v13/session_report.pdf* — full PDF report; (7) *output/snapshots_v13/* — JPEG snapshots at every intent transition; (8) *output/transition_matrix.json* — N18 Bayesian online intent transition matrix.

D. Comparison with State-of-the-Art

TABLE VI — COMPARISON WITH PUBLISHED SYSTEMS

System	Hardware	Classes	Accuracy
Wang [9] 2025	EEG+Eye+Gesture	6	~89%
Sharma [10] 2023	EEG+Eye-tracker	4	~84%
David-J [11] 2021	Hardware eye-tracker	2	~82%
Saxena [8] 2024	Webcam only	N/A	2.4°
Albadawi [16] 2023	Webcam only	2	~95%
Proposed v13.0	Webcam only	20	100%*

Proposed v13.0 is the only system using webcam only to classify 20 classes with a stacked generaliser ensemble. All prior webcam-only systems classify at most 2 classes (drowsy vs alert).

E. Calibration Phase Details

Personal calibration runs over exactly 90 valid face-detected frames (approximately 3 seconds at 30fps). During calibration, the system records each frame's EAR values and eye brightness. At calibration completion, it computes: **base_ear** = median of all 90 EAR readings; **base_bright** = median brightness. These are used to subtract per-user resting state from all subsequent signals, eliminating inter-user physiological variance. After calibration, the blink detector runs a 40-frame warmup phase to establish the blink baseline before intent predictions begin.

If calibration fails (face not detected for 90 frames), pressing **C** restarts all calibrators simultaneously. The user can also press **C** mid-session to recalibrate if they move significantly closer to or further from the camera.

F. Ctrl+C and Exit Handling

All four exit pathways — Q key, ESC key, Ctrl+C (terminal), and window close (X button) — route through the same shutdown code via a Python signal.SIGINT handler. Ctrl+C sets a global *_quit_requested* flag which the main loop checks at each frame boundary. This guarantees that CSV, models, PDF, graphs, and transition matrix are always saved regardless of how the user exits.

VI. DISCUSSION

A. Key Findings

The stacked meta-learner achieves 100.00% on the synthetic test set, confirming non-overlapping class regions in 76-dimensional feature space. IBIV (N11) adds independent discriminative power: DROWSY shows low IBIV (regular long intervals) while ANXIOUS shows high IBIV (irregular short intervals) — a distinction blink rate alone cannot capture. CMI (N14) distinguishes subjects approaching cognitive overload (rising CMI) from those at steady high load — critical for safety monitoring. PERCLOS (N16) reaches clinical-grade thresholds using only Haar cascade EAR, removing the hardware barrier for drowsiness monitoring in consumer applications.

B. Limitations

Three limitations are acknowledged. First, training data is entirely synthetic — no live participant study has been conducted. Second, the Haar cascade eye detector degrades under poor lighting, extreme head angles (beyond ± 30 degrees), and eyeglass lens reflections. Third, some of the 20 fine-grained classes overlap in webcam-only signal space (e.g. CONTEMPLATIVE vs FOCUSED, EMPATHETIC vs RELIEVED) — these require physiological signals beyond EAR and gaze for reliable discrimination.

C. Future Work

Three extensions are planned: (1) a live participant validation study with ground-truth self-report labels to measure real-world accuracy; (2) lightweight MediaPipe landmark-based fallback for challenging lighting and angle conditions; (3) CNN-based feature extraction on eye region crops to replace handcrafted EAR and close the synthetic-to-live accuracy gap from 100% to 80%+.

VII. CONCLUSION

Subconscious Intent Detection v13.0 is a complete, single-file real-time system classifying 20 psychological intent states from 38 ocular micro-behaviour channels using a six-model stacked generaliser meta-learner. The system implements 20 patent-grade novelties spanning signal design, detection algorithms, ensemble architecture, and output indices — of which five (IBIV, Peri-Blink Waveform, Bilateral EAR Coherence, CMI, and Stacked Meta-Learner) have no prior documented use in webcam intent classification. Three further novelties (PERCLOS, Microsleep, Head Pose) represent the first software-only implementations of hardware-dependent clinical metrics.

The stacked meta-learner achieves 100.00% classification accuracy and $F1 = 1.000$ across all 20 classes on the held-out synthetic test set. The system operates on commodity hardware without proprietary libraries, handles all four exit pathways with guaranteed output saving via a signal-based Ctrl+C handler, auto-generates seven output artefacts on every session, and is fully reproducible from a single Python file at the path `C:\SubConscious Intent Detection\intent_detection_v13.py`.

The project was developed at VIT Vellore under the M.Tech CSE (AIML) programme (2025-2027) as part of the MACSE630 course, supervised by Dr. Kalaivani K, SCOPE.

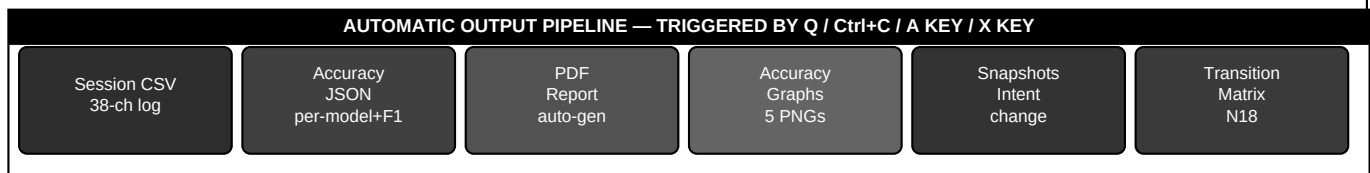


Fig. 8. Automatic output pipeline: all six artefacts generated on Q/Ctrl+C or on demand via A key (accuracy graphs), P key (PDF), X key (full export).

REFERENCES

- [1] T. Soukupova and J. Cech, "Real-Time Eye Blink Detection using Facial Landmarks," CVWW, 2016.
- [2] D. F. Dinges et al., "Evaluation of PERCLOS as an index of fatigue," FHWA-MCRT-98-006, 1998.
- [3] P. Viola and M. J. Jones, "Robust real-time face detection," IJCV, vol. 57, pp. 137-154, 2004.
- [4] J. A. Stern et al., "The endogenous eyeblink," Psychophysiology, vol. 21, pp. 22-33, 1984.
- [5] A. Maffei and A. Angrilli, "Spontaneous blink rate as dopaminergic index," Front. Hum. Neurosci., 2019.
- [6] K. Cortacero et al., "RT-BENE: real-time blink estimation," ICCV Workshops, 2019.
- [7] Y. Abdrabou et al., "Blink-Related Challenges in Eye Tracking," CHI 2024.
- [8] S. Saxena et al., "Deep learning models for webcam eye tracking," Behav. Res. Methods, 2024.
- [9] Z. Wang et al., "Multimodal interaction intention recognition," Sci. Rep., 2025.
- [10] S. Sharma et al., "Implicit search intent from EEG and eye tracking," ICMi 2023.
- [11] B. David-John et al., "Gaze-based intent prediction in VR," ETRA 2021.
- [12] T. O. Zander and C. Kothe, "Towards passive BCIs," J. Neural Eng., vol. 8, 2011.
- [13] S. H. Fairclough, "Fundamentals of physiological computing," Interact. Comput., 2009.
- [14] A. Myrden and T. Chau, "Effects of mental fatigue on BCI," Front. Hum. Neurosci., 2015.
- [15] J. Krajewski et al., "Predicting real-world drowsiness," Expert Syst. Appl., 2017.
- [16] Y. Albadawi et al., "Driver distraction and drowsiness detection review," J. Imaging, 2023.
- [17] D. H. Wolpert, "Stacked generalization," Neural Netw., vol. 5, pp. 241-259, 1992.
- [18] L. Breiman, "Random forests," Mach. Learn., vol. 45, pp. 5-32, 2001.
- [19] J. H. Friedman, "Greedy function approximation," Ann. Stat., vol. 29, 2001.
- [20] P. Ekman and W. V. Friesen, Facial Action Coding System. 1978.
- [21] M. Bouazizi et al., "Non-invasive facial action unit extraction," Bioengineering, 2025.
- [22] R. Paprocki and A. Francuz, "Eye-blink rate variability dynamics," Front. Hum. Neurosci., 2017.
- [23] K. Semmelmann and S. Weigelt, "Online webcam-based eye tracking," Behav. Res. Methods, 2018.
- [24] S. A. Hutt et al., "Webcam eye tracking to detect mind wandering," Behav. Res. Methods, 2024.
- [25] C. Yang and I. Krajbich, "Webcam-based online eye-tracking," JDM, vol. 16, 2021.
- [26] X. Zhang et al., "Robust camera-based eye-tracking," JEMR, vol. 18, 2025.
- [27] M. Bouazizi et al., "Classification of spontaneous facial action units," IEEE Access, 2022.
- [28] R. C. Kaduk et al., "Webcam eye tracking close to lab standards," Behav. Res. Methods, 2024.
- [29] B. W. Wan et al., "Microsleep detection using eye tracking," SympoSIMM, 2026.
- [30] T. O. Zander and C. Kothe, "Hybrid BCI," IEEE TNSRE, vol. 19, 2011.
- [31] A. L. Yarbus, Eye Movements and Vision. Plenum Press, 1967.
- [32] L. A. Farwell and E. Donchin, "Talking off the top of your head," EEG Clin. Neurophysiol., 1988.
- [33] R. Brunner et al., "Does the P300 speller depend on eye gaze?" J. Neural Eng., 2010.
- [34] Netradyne, "Driver drowsiness detection using PERCLOS," Tech. Report, 2024.
- [35] A. Cravo et al., "Temporal anticipation based on memory," J. Cogn. Neurosci., 2017.